KARTHIK NAYAK

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EDUCATION

University of Maryland - College Park | M.Eng. in Software Engineering – 3.85/4.0 — Jan 2024 – Aug 2025

- **Relevant course-work:**
  - Real Time Operating Systems, Reverse Software Engineering, AI in Software Systems, SW Requirements, Design and Testing.

KLS Gogte Institute of Technology - India | B.E. in Electronics & Communication – 9.17/10 — Aug 2015 – May 2019

SKILLS

C | C++ | Python | Assembly | GTest | pytest

FreeRTOS | Embedded | UART, I2C, SPI | Memory Management, Bootloader and Boot Sequence | AUTOSAR | ASPICE

NumPy | Pandas | Matplotlib | TensorFlow | Keras | OpenCV

Object Oriented Programming | PlantUML | Git | JIRA | Agile | Software Requirements | Software Testing

TECHNICAL EXPERIENCE

Bosch Global Software Technologies, Bangalore | Senior Engineer — Oct 2022 - Jan 2024

- Worked as a software developer in the **radar perception** team of the **Advanced Driver Assistance Systems (ADAS) unit**.
- Developed **C++ controllers in a RTOS framework** to capture and **process radar data in real-time** to ensure efficient data flow into a neural network model, which predicted whether the **radar was obstructed by snow, mud or dirt**.
- **Ported, optimized and unit tested** trained neural network models from the Keras framework for **deployment on micro-controllers**.
- **Researched deep learning concepts** such as cyclic learning rate, calibration of neural network models input data pipelines to enhance model accuracy while reducing training time.

Continental Automotive Components, Bangalore | Engineer — Oct 2019 - Oct 2022

- Worked as a software developer in the **Rear View Camera** team of the **ADAS unit**.
- Implemented **AUTOSAR compliant C Tasks in EB Tresos** RTOS framework enabling seamless switching of camera outputs during reverse parking maneuvers. In order to improve driver visibility and decision-making, this entailed providing capabilities for dynamic transitions between **camera feed with and without guiding overlays**. Worked in conjunction with cross-functional teams to incorporate these features into the overall framework and performing extensive testing to guarantee their dependability in a variety of scenarios.
- Worked in the task force for **runtime optimization** in order to meet client requirements and drastically lower total system load. **Minimized needless lost ECU cycles** by switching the application from exisitng **polling-based to an event-driven** architecture. **Optimized task scheduling and prioritization** in order to minimize memory usage. System load was thus decreased from **131.5pc to 51pc**, which **enhanced efficiency and performance**.
- Implemented a **UART driver** to log **debug logs** such as memory load, exceptions, and state transitions, which enabled quick debugging and accelerated development.
- Implemented an **I2C driver** to read **temperature data from SoC**, needed as part of diagnostic data.
- Used **PlantUML** to create extensive **state, activity and sequence diagrams** that helped with design clarity and documentation by graphically representing system workflows and interactions.
- Finalized modular software interfaces in **collaboration with software architects and component owners**, examining design specifications and data flow to guarantee smooth component. **integration and compatibility**.

Dept. of Computer Science, IIT Bombay | Summer Intern — May 2018 – July 2018

- Worked on **auto-tuning of controller for drone** which resulted in faster, easier and efficient tuning of **PID controller** that controlled drone flight.
- Developed **ROS py nodes** to auto compute the **Kp, Ki, Kd constants** of the PID controller.
- Developed **path planner for the drone to hover and fly** in a given indoor area. The drone was localized via whycon marker and overhead camera.

AWARDS AND CERTIFICATIONS

- **Team of the quarter Award**: Issued by Continental Automotive Components India Pvt. Ltd., received this award as part of Star of the quarter (Team) award for **Runtime Optimization of Rear View Camera System**.
- **Programming in Modern C++** - NPTEL